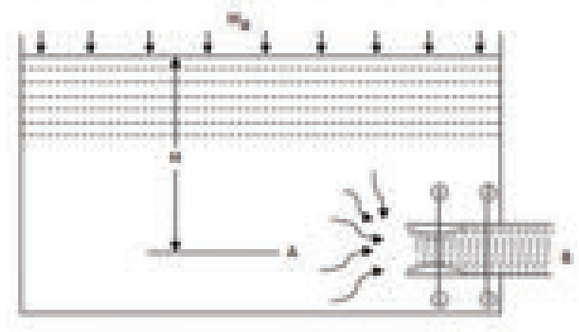
7.4.8 Types of mouthpieces

Mouthpieces are classified as below

1. According to location
   - a. Internal mouthpiece



   - b. External mouthpiece



2. According to shape
   - a. Cylindrical mouthpiece
   - b. Convergent mouthpiece
   - c. Divergent mouthpiece
   - d. Convergent - divergent mouthpiece
3. According to nature of discharge
   - a. Running full
   - b. Running free

7.5 Flow Through Pipes

7.5.1 Definition

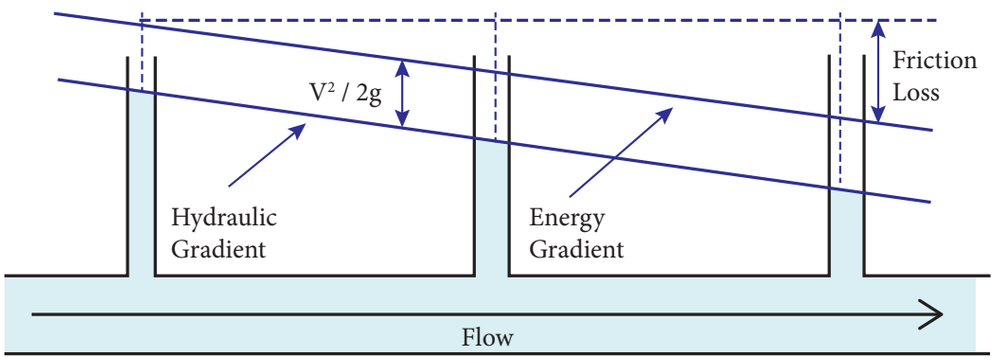
A pipe is a conduit, generally of circular cross section, used to carry liquid under pressure. When the pipe is running full, the flow is under pressure. This kind of flow is known as pipe flow. But, if the pipe is not running full, the flow is not under pressure. In such a case, the atmospheric pressure exist inside the pipe. This kind of flow is termed as channel flow.

7.5.2 Hydraulic gradient line

If pressure heads $\frac{P}{\gamma}$ of a flowing liquid in a pipe be plotted as vertical ordinates on the centerline of the pipe, then the line joining the tops of such ordinates is knows as hydraulic gradient line (H.G.L).

7.5.3 Total energy line

If the sum of pressure heads and velocity heads $\left(\frac{P}{\gamma}+\frac{v^2}{2g}\right)$ of a liquid flowing in a pipe be plotted as vertical ordinates on the center line of the pipe, then the line



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